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YUVA INTERN PROGRAM

SQL DATA ANALYST - AGRIBUSINESS

Week 4 Project Report
Reporting and Visualization for Agribusiness Insights

Dataset:

Agribusiness Database - MySQL

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Introduction

Agribusiness operations generate data related to crop production, cultivated area, yield, farms, seasons and operational costs. Analysing this data can help a business understand how its operations are performing and where improvements may be possible.

In the previous stages of this project, the agribusiness data was cleaned, organised into a relational database and analysed using SQL. The analysis produced information about crop production, yield efficiency, farm performance, yearly and seasonal trends and operational costs.

This final stage focuses on presenting these findings in a clear business report. The aim is to bring together the SQL results, visualisations and business insights so that the information can be understood by non-technical stakeholders without requiring detailed knowledge of SQL.

The report revisits the key findings from the earlier analysis and presents them using tables, charts and supporting SQL queries. It also explains why specific visualisation formats were selected and how they help communicate the results. The findings are then used to develop practical recommendations for improving agribusiness operations.

Overall, this report brings together the complete analytical journey, from SQL-based data extraction to visual presentation and business recommendations.

Business Scenario

The agribusiness company manages multiple farms that grow different crops across different seasons and years. The business maintains records of cultivated area, production quantity, crop yield and operational costs.

Management needs a clear view of this information to understand which crops and farms are performing well and where operational improvements may be needed. The earlier SQL analysis showed differences in production, yield and cost across crops, farms and seasons.

For example, wheat recorded the highest total production at 1,175 units in the crop-level analysis while tomato recorded the highest yield per acre at 9.00 units. Labour was also the largest operational cost at 96,500. These differences show why production alone cannot be used to judge overall performance.

The purpose of this final report is to bring these findings together and present them in a clear format for business stakeholders. SQL results will be supported with tables and visualisations so that important trends and differences can be understood quickly. The findings will then be used to suggest practical improvements in production efficiency, cost management and performance monitoring.

Methodology

The analysis was carried out using the agribusiness relational database developed in the previous stages of the project. The database contains information about farms, crops, seasons, production records and operational costs.

The analysis followed these main steps:

Data Review

The available tables and relationships were reviewed to identify the information required for the final business report.

SQL Data Extraction

SQL queries were used to extract production, yield and cost information from the relevant tables. Joins were used where information from multiple tables was required. Aggregate functions such as sum() and avg() were used to calculate performance measures.

KPI Calculation

Key measures such as total production, average yield, yield per acre and cost per unit were calculated to compare different areas of the business.

Result Interpretation

The SQL outputs were reviewed to identify important differences and trends. Actual values from the query results were used to support the findings rather than relying only on general observations.

Visualization

Charts and tables were planned based on the type of information being presented. Column charts were used for category comparisons while tables were used where exact values were important.

Business Interpretation

The results were converted into simple business insights and recommendations. The final report presents the technical results in a format that can be understood by non-technical stakeholders.

SQL Data Extraction

SQL was used to extract the key performance information required for the final business report. The queries developed during the earlier analysis were revisited and selected based on their relevance to business reporting.

The main areas extracted were crop production, crop yield, farm performance, yearly performance, seasonal performance and operational costs. Additional calculations were used to compare the cost of producing different crops.

The extracted results provide the numerical evidence used throughout this report. For example, the crop production query showed that wheat had the highest total production at 1,175 units while the yield analysis showed that tomato had the highest yield per acre at 9.00 units. The cost analysis showed that labour was the largest operational cost at 96,500.

The SQL queries were also reviewed to make sure that each result could be directly connected to a business question. The results were then organised into tables and visualisations to make the information easier for stakeholders to interpret.

The following sections present the main SQL queries along with their purpose, results and business interpretation.

Key Results

The SQL analysis produced several results that are useful for understanding agribusiness performance. The results below focus on production, yield, farm performance, yearly and seasonal trends and operational costs.

Each analysis includes the SQL query used to extract the result, an explanation of its purpose and the key values obtained. These results form the basis for the visualisations and business insights presented later in the report.

Production Performance by Crop

Business Question:

Which crops have the highest total production?

SQL Query:

```
select
    c.crop_name,
    sum(p.production_quantity) as total_production
from production p
join crop c
    on p.crop_id = c.crop_id
group by c.crop_name
order by total_production desc;
```

Purpose:

This query calculates the total production for each crop by adding the production quantities recorded for that crop. The **join** connects the production records with the crop names so that the results can be presented in a readable form.

Key Results:

Crop	Total Production
Wheat	1,175.00

Rice	905.00
Sugarcane	640.00
Maize	520.00
Tomato	495.00
Potato	275.00
Onion	240.00
Soybean	232.00
Cotton	230.00
Mustard	97.50

	crop_name	total_production
►	Wheat	1175.00
	Rice	905.00
	Sugarcane	640.00
	Maize	520.00
	Tomato	495.00
	Potato	275.00
	Onion	240.00
	Soybean	232.00
	Cotton	230.00
	Mustard	97.50

Interpretation:

Wheat recorded the highest total production at 1,175 units followed by rice at 905 units and sugarcane at 640 units. Mustard had the lowest production at 97.50 units. The difference shows that production levels vary considerably across crops. However, total production alone does not show how efficiently the cultivated area was used, so yield per acre is also considered in the next analysis.

Yield Performance by Crop

Business Question:

Which crops produce more output from the cultivated area?

SQL Query:

```
select
    c.crop_name,
    sum(p.production_quantity) as total_production,
    sum(p.cultivated_area) as total_area,
    round(sum(p.production_quantity) / sum(p.cultivated_area), 2) as yield_per_acre
from production p
join crop c
    on p.crop_id = c.crop_id
group by c.crop_name
order by yield_per_acre desc;
```

Purpose:

This query compares production with the total cultivated area for each crop. The calculated yield per acre shows how much production is obtained from each unit of cultivated land.

Key Results:

Crop	Total Production	Total Area	Yield per Acre
Tomato	495.00	55.00	9.00
Sugarcane	640.00	80.00	8.00
Onion	240.00	30.00	8.00

Rice	905.00	175.00	5.17
Potato	275.00	55.00	5.00
Maize	520.00	110.00	4.73
Wheat	1,175.00	270.00	4.35
Soybean	232.00	110.00	2.11
Cotton	230.00	115.00	2.00
Mustard	97.50	65.00	1.50

	crop_name	total_production	total_area	yield_per_acre
►	Tomato	495.00	55.00	9.00
	Sugarcane	640.00	80.00	8.00
	Onion	240.00	30.00	8.00
	Rice	905.00	175.00	5.17
	Potato	275.00	55.00	5.00
	Maize	520.00	110.00	4.73
	Wheat	1175.00	270.00	4.35
	Soybean	232.00	110.00	2.11
	Cotton	230.00	115.00	2.00
	Mustard	97.50	65.00	1.50

Interpretation:

Tomato had the highest yield per acre at 9.00 units while sugarcane and onion recorded 8.00 units each. Mustard had the lowest yield at 1.50 units per acre. Wheat had the highest total production at 1,175 units but its yield per acre was 4.35. This shows that high total production does not necessarily mean high land-use efficiency.

Farm Performance

Business Question:

How does production performance differ between farms?

SQL Query:

```
select
    f.farm_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield
from production p
join farm f
    on p.farm_id = f.farm_id
group by f.farm_name
order by total_production desc;
```

Purpose:

This query compares farms using total production and average yield. Total production shows the overall output of each farm while average yield provides an additional measure of production performance.

Key Results:

Farm	Total Production	Average Yield
Sunrise Agro Farm	1,340.00	5.83
Ganga Plains Farm	745.00	4.83
Green Valley Farm	635.00	3.33
Haryana Fields Farm	562.50	5.33

Golden Harvest Farm	555.00	4.00
Nashik Harvest Farm	465.00	5.50
Agra Green Farm	375.00	5.00
Deccan Agro Farm	132.00	2.20

	farm_name	total_production	average_yield
►	Sunrise Agro Farm	1340.00	5.83
	Ganga Plains Farm	745.00	4.83
	Green Valley Farm	635.00	3.33
	Haryana Fields Farm	562.50	5.33
	Golden Harvest Farm	555.00	4.00
	Nashik Harvest Farm	465.00	5.50
	Agra Green Farm	375.00	5.00
	Deccan Agro Farm	132.00	2.20

Interpretation:

Sunrise Agro Farm recorded the highest total production at 1,340 units with an average yield of 5.83. Nashik Harvest Farm produced 465 units but had an average yield of 5.50, which is higher than several farms with greater total production. Deccan Agro Farm had the lowest total production at 132 units and the lowest average yield at 2.20. This shows why both production and yield should be considered when comparing farm performance.

Yearly Performance

Business Question:

How did production and average yield change between the two years?

SQL Query:

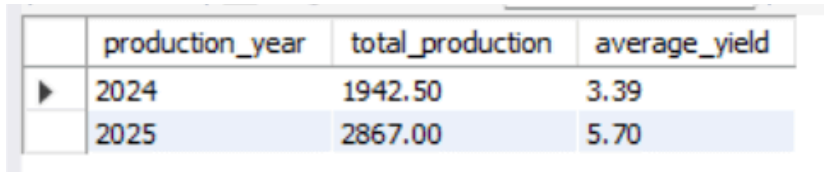
```
select
    production_year,
    sum(production_quantity) as total_production,
    round(avg(yield_quantity), 2) as average_yield
from production
group by production_year
order by production_year;
```

Purpose:

This query groups production records by year to compare total production and average yield over time. It helps identify whether overall performance increased or decreased between the available years.

Key Results:

Production Year	Total Production	Average Yield
2024	1,942.50	3.39
2025	2,867.00	5.70



	production_year	total_production	average_yield
▶	2024	1942.50	3.39
	2025	2867.00	5.70

Interpretation:

Total production increased from 1,942.50 units in 2024 to 2,867.00 units in 2025. Average yield also increased from 3.39 to 5.70. This indicates an improvement in both overall production and average yield in 2025 compared with 2024.

Seasonal Performance

Business Question:

How does production and yield differ across growing seasons?

SQL Query:

```
select
    s.season_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield
from production p
join season s
    on p.season_id = s.season_id
group by s.season_name
order by total_production desc;
```

Purpose:

This query compares total production and average yield across seasons. It helps identify seasons with higher overall production and seasons that achieve better average yield.

Key Results:

Season	Total Production	Average Yield
Kharif	2,527.00	4.12
Rabi	1,547.50	3.71
Summer	735.00	8.67

	season_name	total_production	average_yield
►	Kharif	2527.00	4.12
	Rabi	1547.50	3.71
	Summer	735.00	8.67

Interpretation:

Kharif recorded the highest total production at 2,527 units. However, Summer had the highest average yield at 8.67 units despite having lower total production of 735 units. This difference shows that a season with lower overall production can still have better yield efficiency.

Operational Cost Analysis

Business Question:

Which operational activities contribute the most to the overall cost?

SQL Query:

```
select
    cost_type,
    sum(amount) as total_cost
from cost
group by cost_type
order by total_cost desc;
```

Purpose:

This query groups the recorded costs by cost type and calculates the total amount spent on each category. It helps identify which operational activities contribute most to the overall cost structure.

Key Results:

Cost Type	Total Cost
Labour	96,500.00
Machinery	30,000.00
Irrigation	29,300.00
Transportation	12,600.00

	cost_type	total_cost
►	Labour	96500.00
	Machinery	30000.00
	Irrigation	29300.00
	Transportation	12600.00

Interpretation:

Labour was the largest operational cost at 96,500, followed by machinery at 30,000 and irrigation at 29,300. Transportation had the lowest cost at 12,600. Labour accounted for about 57.3% of the total operational cost, making it an important area for cost monitoring. The business can review labour requirements and work allocation to identify opportunities for reducing unnecessary expenses without affecting farm operations.

Cost per Unit of Production

Business Question:

How much does it cost to produce one unit for each production record?

SQL Query:

```
select
    p.production_id,
    c.crop_name,
    p.production_quantity,
    sum(co.amount) as total_cost,
    round(sum(co.amount) / p.production_quantity, 2) as cost_per_unit
from production p
join crop c
    on p.crop_id = c.crop_id
join cost co
    on p.production_id = co.production_id
group by
    p.production_id,
    c.crop_name,
    p.production_quantity
order by cost_per_unit desc;
```

Purpose:

This query combines production and cost information to calculate the cost required to produce one unit for each production record. This helps identify production records with relatively high or low cost efficiency.

Key Results:

Production ID	Crop	Production	Total Cost	Cost per Unit
2	Mustard	45.00	4,500.00	100.00
11	Mustard	52.50	4,800.00	91.43
5	Cotton	80.00	7,000.00	87.50
18	Cotton	150.00	12,000.00	80.00
14	Soybean	100.00	6,500.00	65.00
20	Soybean	132.00	7,500.00	56.82
1	Wheat	320.00	14,000.00	43.75
6	Potato	125.00	5,000.00	40.00
7	Wheat	450.00	16,500.00	36.67
13	Wheat	405.00	14,500.00	35.80
17	Potato	150.00	5,100.00	34.00
3	Maize	270.00	8,500.00	31.48

4	Rice	350.00	10,800.00	30.86
12	Tomato	180.00	5,200.00	28.89
8	Maize	250.00	7,000.00	28.00
16	Rice	225.00	6,200.00	27.56
10	Rice	330.00	9,000.00	27.27
15	Onion	240.00	5,300.00	22.08
19	Tomato	315.00	6,500.00	20.63
9	Sugarcane	640.00	12,500.00	19.53

	production_id	crop_name	production_quantity	total_cost	cost_per_unit
►	2	Mustard	45.00	4500.00	100.00
	11	Mustard	52.50	4800.00	91.43
	5	Cotton	80.00	7000.00	87.50
	18	Cotton	150.00	12000.00	80.00
	14	Soybean	100.00	6500.00	65.00
	20	Soybean	132.00	7500.00	56.82
	1	Wheat	320.00	14000.00	43.75
	6	Potato	125.00	5000.00	40.00
	7	Wheat	450.00	16500.00	36.67
	13	Wheat	405.00	14500.00	35.80
	17	Potato	150.00	5100.00	34.00
	3	Maize	270.00	8500.00	31.48
	4	Rice	350.00	10800.00	30.86
	12	Tomato	180.00	5200.00	28.89
	8	Maize	250.00	7000.00	28.00
	16	Rice	225.00	6200.00	27.56
	10	Rice	330.00	9000.00	27.27
	15	Onion	240.00	5300.00	22.08
	19	Tomato	315.00	6500.00	20.63

Interpretation:

Production ID 2 for mustard had the highest cost per unit at 100.00 while production ID 9 for sugarcane had the lowest at 19.53. The results also show that the same crop can have different cost levels across production records. This suggests that production conditions, input use or operational activities can affect cost efficiency.

Overall Crop Performance

Business Question:

How do crops compare when production, yield and cost are considered together?

SQL Query:

```
select
    c.crop_name,
    sum(p.production_quantity) as total_production,
    round(avg(p.yield_quantity), 2) as average_yield,
    sum(co.amount) as total_cost,
    round(sum(co.amount) / sum(p.production_quantity), 2) as cost_per_unit
from production p
join crop c
    on p.crop_id = c.crop_id
join cost co
    on p.production_id = co.production_id
group by c.crop_name
order by cost_per_unit;
```

Purpose:

This query combines production, yield and cost information to provide an overall view of crop performance. Looking at these measures together gives a broader understanding than using production or yield alone.

Key Results:

Crop	Total Production	Average Yield	Total Cost	Cost per Unit
Sugarcane	1,920.00	8.00	12,500.00	6.51

Onion	480.00	8.00	5,300.00	11.04
Tomato	990.00	9.00	11,700.00	11.82
Rice	2,160.00	5.14	26,000.00	12.04
Wheat	3,525.00	4.33	45,000.00	12.77
Maize	1,040.00	4.75	15,500.00	14.90
Potato	550.00	5.00	10,100.00	18.36
Soybean	464.00	2.10	14,000.00	30.17
Cotton	610.00	2.00	19,000.00	31.15
Mustard	195.00	1.50	9,300.00	47.69

	crop_name	total_production	average_yield	total_cost	cost_per_unit
►	Sugarcane	1920.00	8.00	12500.00	6.51
	Onion	480.00	8.00	5300.00	11.04
	Tomato	990.00	9.00	11700.00	11.82
	Rice	2160.00	5.14	26000.00	12.04
	Wheat	3525.00	4.33	45000.00	12.77
	Maize	1040.00	4.75	15500.00	14.90
	Potato	550.00	5.00	10100.00	18.36
	Soybean	464.00	2.10	14000.00	30.17
	Cotton	610.00	2.00	19000.00	31.15
	Mustard	195.00	1.50	9300.00	47.69

Interpretation:

Sugarcane had the lowest cost per unit at 6.51 while also having a strong average yield of 8.00. Tomato had the highest average yield at 9.00 and a cost per unit of 11.82. Wheat recorded the highest total production at 3,525 units in this combined analysis with a cost per unit of 12.77. Mustard had the highest cost per unit at 47.69 and the lowest average yield at 1.50. These results show that crop performance should be evaluated using production, yield and cost together rather than relying on a single measure.

Visualization Planning

Visualisations were planned to make the SQL results easier to understand for non-technical stakeholders. The chart type for each result was selected based on the type of comparison being presented.

Total Production by Crop

A column chart is used to compare the total production of different crops. A column chart is suitable because the crops are separate categories and their production values can be compared directly. The chart makes differences such as wheat at 1,175 units and mustard at 97.50 units easy to identify.

Yield per Acre by Crop

A column chart is used to compare yield per acre across crops. This makes it easier to identify crops with higher and lower land-use efficiency. For example, tomato has a yield of 9.00 units per acre while mustard has a yield of 1.50 units per acre.

Total Production by Year

A column chart is used to compare production between 2024 and 2025. With only two years in the dataset, this format clearly shows the increase from 1,942.50 units to 2,867.00 units.

Operational Cost by Cost Type

A column chart is used to compare the different operational cost categories. This makes the difference between labour at 96,500 and transportation at 12,600 clearly visible.

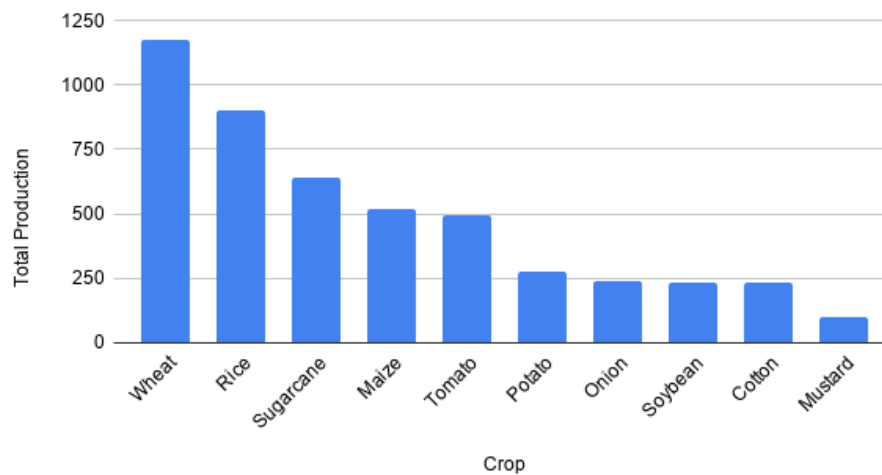
The visualisations support the SQL results by presenting the numerical differences in a simple visual format. They allow stakeholders to identify important patterns quickly without needing to examine every SQL result in detail.

Visual Analysis

The visualisations below present the main findings from the SQL analysis. Each chart focuses on a specific business question and helps make the differences in production, yield and costs easier to understand.

Total Production by Crop

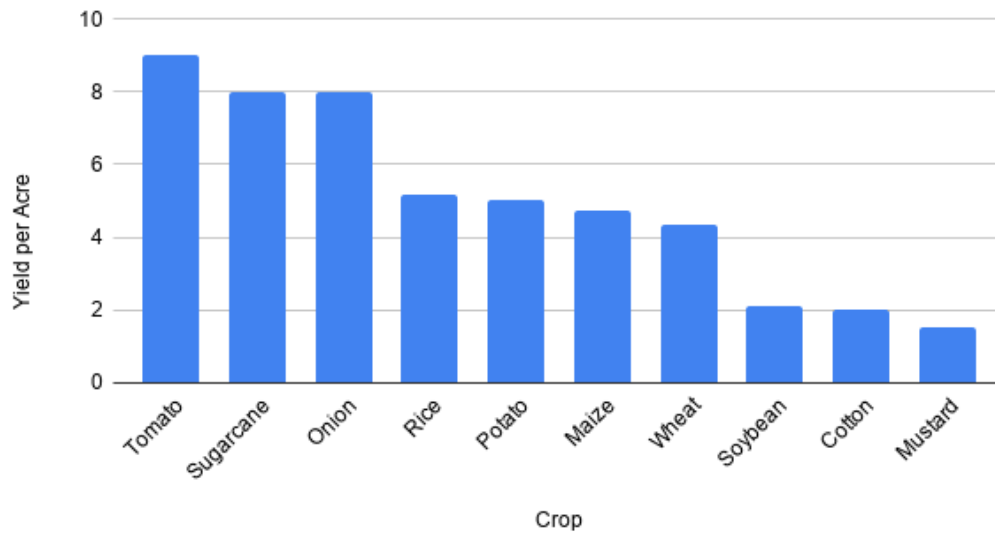
Total Production by Crop



The chart shows clear differences in production across crops. Wheat has the highest total production at 1,175 units, followed by rice at 905 units and sugarcane at 640 units. Mustard has the lowest production at 97.5 units. The large difference between the highest and lowest values shows that production levels vary considerably across crops.

Yield per Acre by Crop

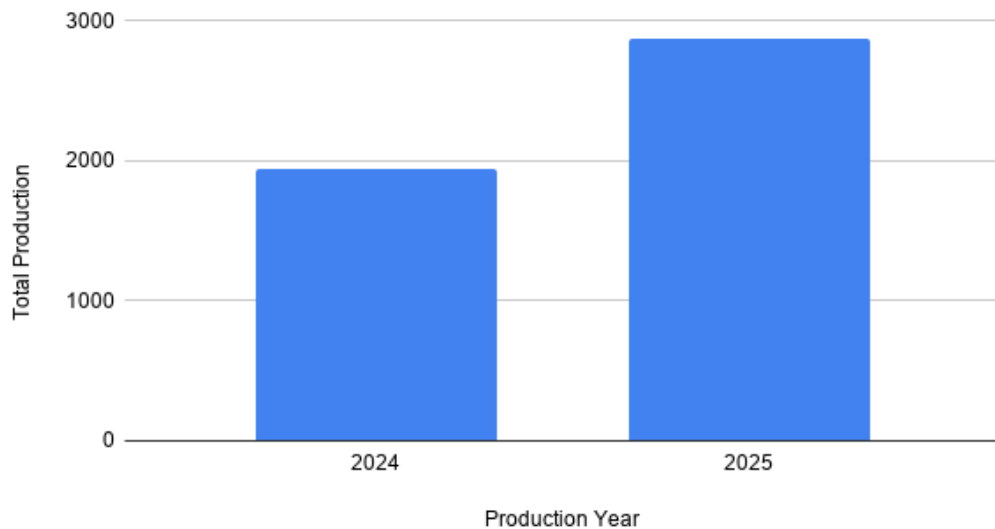
Yield per Acre by Crop



The chart compares how efficiently different crops use cultivated land. Tomato has the highest yield at 9.00 units per acre, while sugarcane and onion both have a yield of 8.00 units per acre. Mustard has the lowest yield at 1.50 units per acre. The comparison shows that total production and land-use efficiency are not always the same.

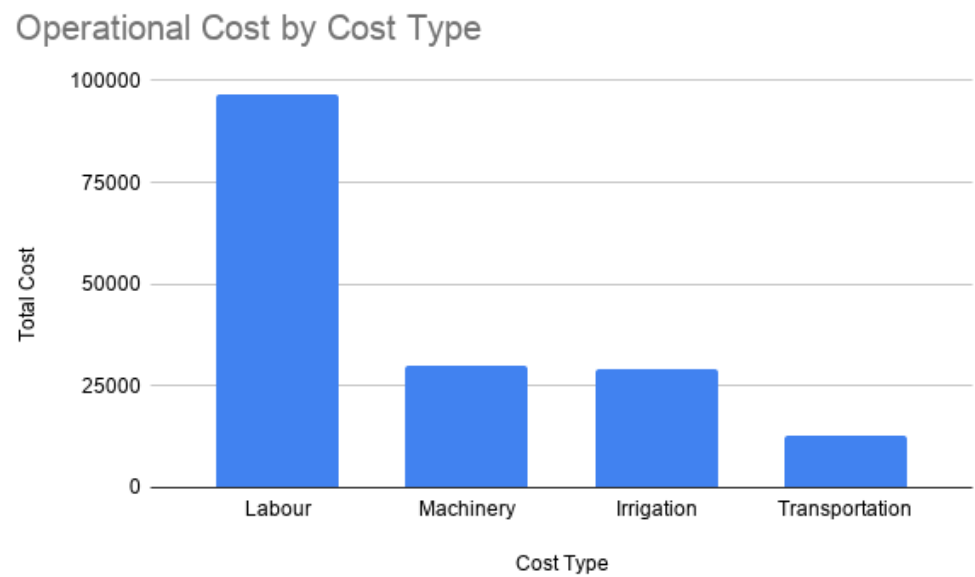
Total Production by Year

Total Production by Year



The chart shows an increase in total production from 1,942.50 units in 2024 to 2,867.00 units in 2025. This represents an increase of 924.50 units, or approximately 47.6%. Average yield also increased from 3.39 to 5.70 units, showing improvement in overall production performance.

Operational Cost by Cost Type



The chart shows that labour is the largest operational cost at 96,500, followed by machinery at 30,000 and irrigation at 29,300. Transportation has the lowest cost at 12,600. Labour therefore represents the main cost area and may require closer monitoring when reviewing operational efficiency.

Business Insights

The combined SQL results and visualisations provide several important insights into the agribusiness operations. Looking at production, yield and costs together gives a clearer picture than using any single measure.

Crop Performance

Wheat has the highest total production at 1,175 units, but tomato has the highest yield per acre at 9.00 units. This shows that a crop with higher total production does not necessarily use land more efficiently. Sugarcane and onion also show strong yield performance at 8.00 units per acre.

Farm Performance

Sunrise Agro Farm has the highest total production at 1,340 units with an average yield of 5.83 units. Nashik Harvest Farm produces 465 units but has an average yield of 5.50 units, showing that a farm with lower total production can still have strong yield performance. Deccan Agro Farm has the lowest total production at 132 units and the lowest average yield at 2.20 units.

Yearly Performance

Total production increased from 1,942.50 units in 2024 to 2,867.00 units in 2025. Average yield also increased from 3.39 to 5.70 units. This shows an improvement in overall production performance between the two years in the dataset.

Seasonal Performance

Kharif has the highest total production at 2,527 units. However, Summer has the highest average yield at 8.67 units despite having lower total production. This shows that total production and yield efficiency can give different views of seasonal performance.

Cost Performance

Labour is the largest operational cost at 96,500, compared with 30,000 for machinery, 29,300 for irrigation and 12,600 for transportation. Labour accounts for approximately 57.3% of the total operational costs in the dataset, making it an important area for cost monitoring.

Overall Crop Performance

When production, yield and cost are considered together, sugarcane has a relatively strong performance in the dataset. It has an average yield of 8.00 units and the lowest overall cost per unit at 6.51. Tomato has the highest average yield at 9.00 units with a cost per unit of 11.82. Mustard has the lowest average yield at 1.50 units and the highest cost per unit at 47.69.

These results show why agribusiness performance should not be evaluated using production alone. Combining production quantity, yield and cost provides a more useful view of operational performance and helps identify areas that may need further review.

Strategic Recommendations

Based on the SQL results and visual analysis, the following recommendations can help improve agribusiness performance.

Review Crop Efficiency

Crops with stronger yield and lower cost per unit should be studied further to understand the practices behind their performance. Sugarcane recorded a yield of 8.00 units per acre and the lowest overall cost per unit at 6.51, while tomato recorded the highest yield at 9.00 units per acre.

Review High-Cost Crops

Mustard requires further review because it recorded the lowest average yield at 1.50 units per acre and the highest overall cost per unit at 47.69. The business can examine its input use, labour requirements and production practices before making decisions about future cultivation.

Monitor Labour Costs

Labour was the largest operational cost at 96,500, accounting for approximately 57.3% of total operational costs. Management should review labour allocation and identify activities where labour requirements can be planned more efficiently without affecting production.

Study High-Performing Farms

Sunrise Agro Farm recorded the highest total production at 1,340 units with an average yield of 5.83 units. The practices used by better-performing farms can be reviewed to identify methods that may be useful for improving performance at other farms.

Track Seasonal Performance

Kharif recorded the highest total production at 2,527 units, while Summer recorded the highest average yield at 8.67 units. Seasonal performance should therefore be monitored using both total production and yield so that management can understand differences in output and efficiency.

Continue Yearly Performance Monitoring

Production increased from 1,942.50 units in 2024 to 2,867.00 units in 2025, while average yield increased from 3.39 to 5.70 units. Similar KPIs should continue to be tracked each year to identify changes in production efficiency and costs over time.

Overall, the recommendations should be used as areas for further review rather than as immediate decisions. Additional operational information can be considered before making major changes to crop selection, farm activities or resource allocation.

Conclusion

This report brought together the SQL analysis, visualisations and business interpretation of the agribusiness data. The analysis examined crop production, yield efficiency, farm performance, yearly and seasonal trends and operational costs.

The results showed that different crops and farms perform differently across the selected measures. Wheat had the highest total production at 1,175 units, while tomato had the highest yield per acre at 9.00 units. Production increased from 1,942.50 units in 2024 to 2,867.00 units in 2025, while Summer recorded the highest average seasonal yield at 8.67 units.

The cost analysis showed that labour was the largest operational cost at 96,500. The overall crop analysis also showed differences in cost efficiency, with sugarcane recording the lowest cost per unit at 6.51 and mustard recording the highest at 47.69.

The visualisations made these differences easier to understand for non-technical stakeholders, while the SQL queries provided the numerical evidence behind the findings. Combining the technical analysis with simple visual reporting helped convert the database results into useful business information.

Overall, this project demonstrated how SQL can be used not only to extract data but also to calculate performance measures, identify patterns and support business decision-making. The findings can be used as a starting point for further analysis and regular performance monitoring in agribusiness operations.